

Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Data Structures and Algorithms [ECE-4070]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Enumerate and explain the properties of an algorithm with an example for the same. (4)
 - A)
 - B) Elaborate and explain the algorithm used for Space complexity. (3)
 - C) Discuss the significance of a transpose matrix in datastructures. (3)
- 2) Convert the infix expression $A+B*C-D/E*H$ into its equivalent postfix expression. (4)
 - A)
 - B) Suggest an algorithm to explain the operations on Circular queue. (3)
 - C) With an example code, explain the insertion operation at the beginning for a Singly linked list. (3)
- 3) Illustrate and explain the methods for constructing an expression tree. (4)
 - A)
 - B) Mention tree traversal operations and explain Inorder traversal with suitable example. (3)
 - C) Prove that the height of AVL tree with 'n' elements is $O(\log n)$ (3)
- 4) With an illustration, explain the adjacency list to represent graphs. (4)
 - A)
 - B) Suggest an algorithm for Linear search considering both Best case and Worst case complexities. (3)
 - C) Discuss the steps of Merge sort algorithm. (3)
- 5) Suggest an algorithm for Merge sort highlighting the significance of the same. (4)

- A)
- B) Discuss the steps involved in heap sort. (3)
- C) Discuss the necessity of Breadth First Search algorithm. (3)

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